Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 14

Double Inclusion, Disproof, and Partitions

Covers Chartrand & Zhang, Sections 2.2 and 2.4

Status of this set

Ungraded practice. Solutions released with it. Attempt everything first — and for the proof problems, write in *sentences*. That is the skill being practised.

Part I: How to Practice This Material

The template, and the two halves

Proving $A = B$

Half one. Let $x \in A$. . . . reasoning . . . Therefore $x \in B$. Hence $A \subseteq B$.
Half two. Let $y \in B$. . . . reasoning . . . Therefore $y \in A$. Hence $B \subseteq A$.
Conclude. Since each is a subset of the other, $A = B$.
The element is *arbitrary*: you assume only that it belongs to the set you started with. That is what makes the conclusion cover every element at once.

Disproving is much cheaper

One specific pair of sets, and one specific element in one side but not the other. Name both. "It doesn't always work" is not a disproof.

Three mistakes to expect

1. **Proving one inclusion and stopping.** Set equality is a biconditional.
2. **Assuming something about x.** If you use a property the arbitrary element need not have, the argument covers only some elements.
3. **Writing symbols with no sentences.** A proof is prose. Chains of implications with no words are not gradeable.

Part II: Practice Problems

Section A — Proving

1. Prove that $A \cap (B - C) = (A \cap B) - C$ for all sets A, B, C.

2. Prove that $A - (A - B) = A \cap B$ for all sets A and B.

Section B — Disproving

3. Each claim below is *false*. Give a counterexample for each: specific sets, and the specific element that fails.

(a) For all sets A and B, $\quad A - B = B - A$.

(b) For all sets A, B, C, $\quad A - (B - C) = (A - B) - C$.

4. Determine whether $(A \cup B) - B = A$ holds for all sets. Prove it or disprove it. Then state and justify what the left side *does* equal.

Section C — Partitions

5. Let $A = \{1, 2, \ldots, 8\}$. Which of these are partitions of A? For those that are not, say exactly which condition fails.

(a) $\{1, 3, 5, 7\},\ \{2, 4, 6, 8\}$

(b) $\{1, 2, 3\},\ \{4, 5\},\ \{5, 6, 7, 8\}$

(c) $\{1, 2\},\ \{3, 4\},\ \{5, 6\},\ \{7\}$

(d) $\{1, 2, 3, 4\},\ \emptyset,\ \{5, 6, 7, 8\}$

Section D — Writing

6. In five to eight complete sentences, explain why proving $A = B$ requires two separate arguments while disproving it requires only one example. Connect your answer to the definition of set equality and to something you learned before this week.

Section E — Challenge (optional)

7. Prove that if $A \subseteq B$, then $A \cup B = B$. Then state and prove the converse.